LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Bus bars: - Bus bar systems.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Bus bars: - Bus bar systems. Block/schematic explanation of bus bar systems, split busbar systems, concept of emergency busbar. Practical on busbar interconnection system.
5. Specific Objectives	13.1.2.1 Knowledge of Static converting equipment, Bus bars, cable terminations, Switches.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on busbar interconnection system. Safety precautions: Wear insulated gloves, ensure power off, verify zero voltage. Instructor supervision: Monitor connections, check for loose terminals, verify circuit continuity.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a bus bar in an aircraft electrical system?
(b) Name two types of bus bar systems used in aviation.
(c) Why is the concept of an emergency bus bar critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Block schematic explanation of bus bar systems • Function of static converting equipment • Cable terminations and switch integration	L & Ds	PPT, IB	15 min
(b)	• Split busbar systems operation • Emergency busbar concept and activation • Power distribution logic in split systems	L & Ds	PPT, PDF	20 min
(c)	• Busbar interconnection system practical • Terminal identification and torque checks • Circuit continuity verification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a split busbar system isolates faults during normal operation.
(b) Describe the role of the emergency bus bar when main power fails.

12. Home Assignment and Follow Up	Prepare a brief note on the routing and ducting requirements for wires connected to bus bars, referencing the next lesson plan on wire types and routing.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Practical and demonstration on Wires and cables: - Types of wires and cables, routing of wires and cables, open loom, ducted loom conduits, cable loom, and co-axial cables. Special purpose cables: - Ignition cables, thermocouple cables.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min